

Unit 7: Rewriting Polynomial Expressions

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The following B.E.S.T Standards will be covered in this unit:

- MA.912.AR.1.7** Rewrite a polynomial expression as a product of polynomials over the real-number system.
- *Clarification 1: Within the Algebra 1 course, polynomial expressions are limited to 4 or fewer terms with integer coefficients.*

Unit 7, Lesson 1: Factoring and the Distributive Property



Benchmark

MA.912.AR.1.7 Rewrite a polynomial expression as a product of polynomials over the real number system.



Learning Targets

- ☐ I can rewrite a polynomial using the distributive property by factoring out a common factor.
- ☐ I can determine the greatest common monomial factor.



7.1.1 Warm-Up: Electric Motorcycle Engineer

1. Marc Fenigstein says, “Open-ended problems are the problems students will face in the real world ... Forcing students to learn how to function in ambiguity and find their own answers ... and that there isn’t necessarily any one right answer, but there might be better or worse answers ... I think that is really valuable. Don’t be afraid to find your own path ... You’re not going to get it right on the first try, but you can make almost any job your dream job by looking for opportunities to do more, to go beyond ...”

a. What stands out to you the most in this statement?

b. Why does that part of the statement stand out to you?



7.1.2 Exploration: The Distributive Property in Factoring

1. Multiply $9(10 + 6)$ using the distributive property.
2. Discuss with your partner how $9(10 + 6)$ shows that 9 and 16 are factors of 144.
3. Discuss with your partner how multiplying $8(287)$ can be done more easily using the distributive property. Explain or show your work.
4. Rewrite the expression $-7(2x - 5)$ using the distributive property.

5. Complete the table.

Expression	Factors	Product
$9(10 + 6)$	9 and 16	144
$-7(2x - 5)$	-7 and $2x - 5$	
$(8 - 4x)10$		
$-12(7 - 3x)$		
$2(6x^2 + 8x)$		
$4x(3x + 4)$		
$(6x + 8) \cdot 2x$		

6. What do you notice about the product of each of the expressions $2(6x^2 + 8x)$, $4x(3x + 4)$, and $(6x + 8) \cdot 2x$?

Expressions are **equivalent** if they have the same value no matter what number is substituted in for each of the variables.

7. The expression $14x^2 + 28x^3$ is the product of each of the following expressions.

$$2(7x^2 + 14x^3)$$

$$x^2(14 + 28x)$$

$$7x^2(2 + 4x)$$

List the factors for each expression.

Factors of $2(7x^2 + 14x^3)$		Factors of $x^2(14 + 28x)$		Factors of $7x^2(2 + 4x)$	

8. Write two more expressions that are equivalent to $14x^2 + 28x^3$.
9. Compare your expressions with your partner's expression. Write the expressions that you and your partner do not have in common.

10. Complete the table for the factors of $14x^2 + 28x^3$ using two other expressions you created.

Monomial Factor	Binomial Factor
2	$(7x^2 + 14x^3)$
x^2	$(14 + 28x)$
$7x^2$	$(2 + 4x)$



7.1.3 Guided Instruction: Common Monomial Factors

The table in the exploration shows there are several different ways to rewrite the expression $14x^2 + 28x^3$.

1. What monomial factor is the greatest common monomial factor of $14x^2 + 28x^3$?
2. Discuss with your partner how to determine if a monomial factor is the greatest common monomial factor of a given polynomial. Summarize your discussion.
3. Determine the greatest common factor for 24, -36 , and -54 .
4. Determine the greatest common factor for x^7 , x^{10} , and x^{13} .

5. Determine the greatest common factor for x^3y^3 , x^2y^8 , and xy^4 .
6. Work with your partner to complete the table. If an expression does not have one of the greatest common factors listed, write N/A.

Expression	Greatest Common Numeric Factor	Greatest Common Variable Factor	Greatest Common Monomial Factor
$25w^3 - 40w^2 + 60w$			
$81g + 108g^2 - 12g^4$			
$7m^3n^2 + 56m^2n^3 - 21mn$			
$12p^2q + 16q^2 - 64p$			



7.1.4 Guided Practice: Factoring with the Distributive Property

- Which of the following expressions are equivalent to $8a^3 - 98a$? Select all that apply.
 - ☐ $a(8a^2 - 98)$
 - ☐ $2(4a^3 - 49a)$
 - ☐ $2a(4a^2 - 49)$
 - ☐ $2a(4a^3 - 49a)$
 - ☐ $4a(2a^2 - 49a)$
 - ☐ $8a^3(2 - 98a)$
- Write three equivalent expressions for $18x^3 - 9x^2 - 15x$. One of the expressions should include the greatest common monomial factor.
- Write an equivalent expression for $2c^3d^3 + 14c^2d + 18cd$ that uses the greatest common monomial factor.



7.1.5 Your Turn!

- Write three equivalent expressions for $72b^2k^3 - 64b^5k^2 + 32b^8k^6$. One of the expressions should include the greatest common monomial factor.
- Which expressions are equivalent to $48r^3 + 18r^2$? Select all that apply.
 - ☐ $3(16r^3 + 6r^2)$
 - ☐ $6r(8r^2 + 3r)$
 - ☐ $2r^2(24r + 9)$
 - ☐ $4(12r^3 + 4r^2)$
 - ☐ $6r^2(8r + 3)$



7.1.6 Wrap-Up: Ticket Out the Door

1. Write three equivalent expressions for $12a^3b^5 - 16a^2b^2 + 20a^3b$. One of the expressions should include the greatest common monomial factor.

Unit 7, Lesson 2: Factoring a Trinomial with Algebra Tiles



Benchmark

MA.912.AR.1.7 Rewrite a polynomial expression as a product of polynomials over the real number system.



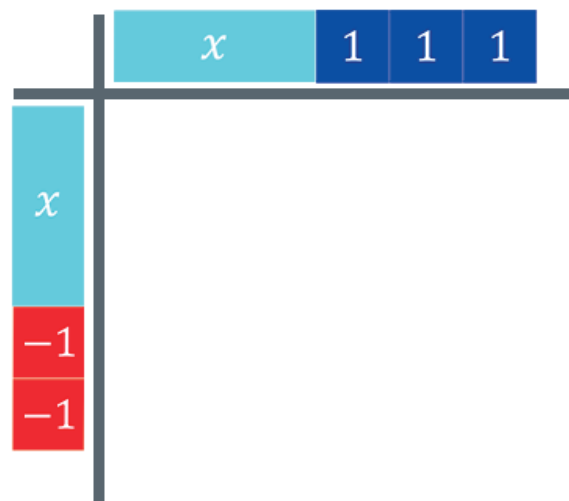
Learning Target

- ☐ I can use algebra tiles to factor a trinomial with a leading coefficient of 1.



7.2.1 Warm-Up: Multiplying Binomials

1. An algebra tile mat is shown with two binomial factors.

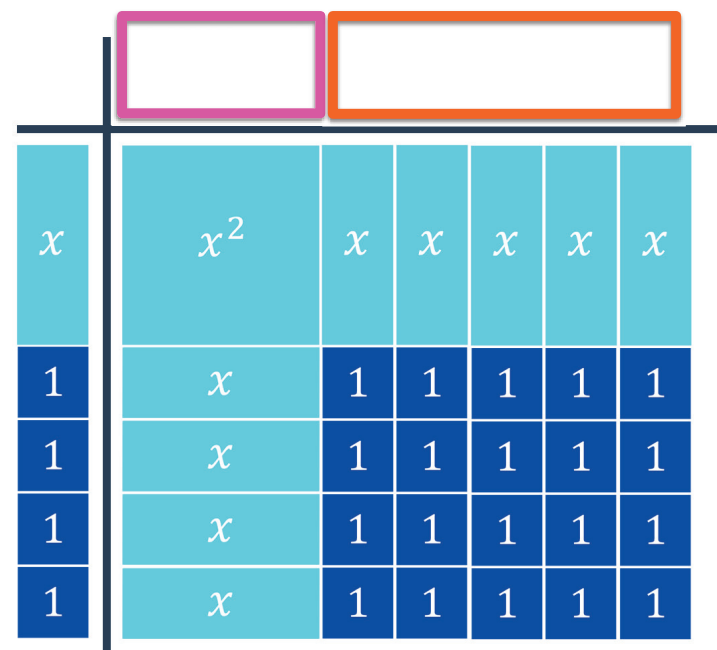


- What are the two binomials represented?
- What is the product of the two binomials?



7.2.2 Exploration: Model Factoring with Algebra Tiles

1. An algebra tile mat is shown, where the product is the trinomial $x^2 + 9x + 20$ and one of its factors is $(x + 4)$.



Diego and Maria Jose both justified how they determined what belonged in the **pink** box in the following ways.

Diego	Maria Jose
I found $\frac{x^2}{x}$ because x^2 is the product of x and something else.	I asked myself what times x would give me x^2 .

a. Discuss with your partner both justifications.

b. Use one of the methods to complete the binomial for the top bar of the algebra tile mat.

	Drawing	Algebraic Representation
Pink Box		x
Orange Box		
Binomial Factor		

2. Discuss with your partner how to determine the two binomial factors when only given the product. Summarize your discussion.

x^2	x	x	x	x	x
x	1	1	1	1	1
x	1	1	1	1	1
x	1	1	1	1	1
x	1	1	1	1	1

The two binomials are the factors of the trinomial $x^2 + 9x + 20$. The process of finding the two binomials is called **factoring**.

3. The algebra tiles shown represent the trinomial $x^2 + 9x + 18$.

x^2	x	x	x
x	1	1	1
x	1	1	1
x	1	1	1
x	1	1	1
x	1	1	1
x	1	1	1

- a. Discuss with your partner what the similarities and differences are between the algebra tiles that represent $x^2 + 9x + 20$ and $x^2 + 9x + 18$. Summarize your discussion.

Similarities	
Differences	

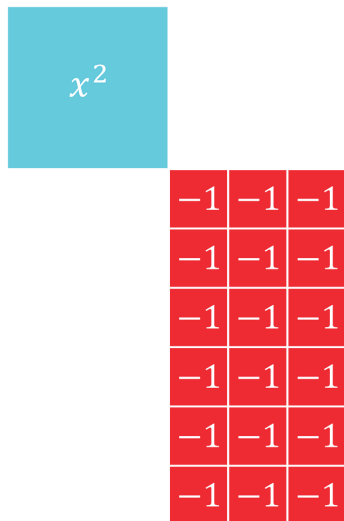
- b. What are the two binomials that are factors of $x^2 + 9x + 18$?

4. The table shows the algebra tile representation for two trinomials. Work with your partner to determine the two binomial factors.

Trinomial	$x^2 - 9x + 18$	$x^2 - 3x - 18$																																																								
Algebra Tiles	<table><tr><td>x^2</td><td>$-x$</td><td>$-x$</td><td>$-x$</td></tr><tr><td>$-x$</td><td>1</td><td>1</td><td>1</td></tr><tr><td>$-x$</td><td>1</td><td>1</td><td>1</td></tr><tr><td>$-x$</td><td>1</td><td>1</td><td>1</td></tr><tr><td>$-x$</td><td>1</td><td>1</td><td>1</td></tr><tr><td>$-x$</td><td>1</td><td>1</td><td>1</td></tr><tr><td>$-x$</td><td>1</td><td>1</td><td>1</td></tr></table>	x^2	$-x$	$-x$	$-x$	$-x$	1	1	1	$-x$	1	1	1	$-x$	1	1	1	$-x$	1	1	1	$-x$	1	1	1	$-x$	1	1	1	<table><tr><td>x^2</td><td>x</td><td>x</td><td>x</td></tr><tr><td>$-x$</td><td>-1</td><td>-1</td><td>-1</td></tr><tr><td>$-x$</td><td>-1</td><td>-1</td><td>-1</td></tr><tr><td>$-x$</td><td>-1</td><td>-1</td><td>-1</td></tr><tr><td>$-x$</td><td>-1</td><td>-1</td><td>-1</td></tr><tr><td>$-x$</td><td>-1</td><td>-1</td><td>-1</td></tr><tr><td>$-x$</td><td>-1</td><td>-1</td><td>-1</td></tr></table>	x^2	x	x	x	$-x$	-1	-1	-1	$-x$	-1	-1	-1	$-x$	-1	-1	-1	$-x$	-1	-1	-1	$-x$	-1	-1	-1	$-x$	-1	-1	-1
	x^2	$-x$	$-x$	$-x$																																																						
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Binomial Factors																																																										

One common attribute of all of the algebra tile representations is that they are all rectangles.

5. Consider the following algebra tile representation.



- Complete the algebra tile representation of a trinomial by drawing and labeling x or $-x$ tiles.
- What is the trinomial that is represented by the completed drawing?
- What are the binomial factors of the trinomial?

6. The algebra tile representation of two different trinomials is shown.

a. Complete the table.

Algebra Tiles	Trinomial	Factors

Each of the trinomials in the table has either twenty-four unit squares, $+1$, or twenty-four negative unit squares, -1 . Also, each trinomial has some combination of four and six $+x$ or $-x$ tiles.

- b. Discuss with your partner how the x -term of each trinomial relates to the four and six ^+x or ^-x tiles.
- c. Discuss with your partner how the constant term, 24 or -24 , of each trinomial relates to the four and six ^+x or ^-x tiles.

7. Consider the trinomial $x^2 - 5x - 24$.

- a. Work with your partner to determine how many ^+x or ^-x tiles are needed for $x^2 - 5x - 24$.

^+x tiles	^-x tiles

- b. Draw a diagram of the algebra tiles that show the trinomial $x^2 - 5x - 24$.

- c. Use your algebra tiles diagram to choose the number of ^+x or ^-x tiles that match your diagram. Then, fill in each blank with the appropriate values.

	x	<u> </u>
x	x^2	☐ $1x$ ☐ $-1x$ ☐ $6x$ ☐ $-6x$ ☐ $2x$ ☐ $-2x$ ☐ $8x$ ☐ $-8x$ ☐ $3x$ ☐ $-3x$ ☐ $12x$ ☐ $-12x$ ☐ $4x$ ☐ $-4x$ ☐ $24x$ ☐ $-24x$
<u> </u>	☐ $1x$ ☐ $-1x$ ☐ $6x$ ☐ $-6x$ ☐ $2x$ ☐ $-2x$ ☐ $8x$ ☐ $-8x$ ☐ $3x$ ☐ $-3x$ ☐ $12x$ ☐ $-12x$ ☐ $4x$ ☐ $-4x$ ☐ $24x$ ☐ $-24x$	-24

- d. Discuss with your partner how using the factors of -24 , and the value of the coefficient of the b -term, -5 , is helpful when factoring the trinomial $x^2 - 5x - 24$. Summarize your discussion.



7.2.3 Guided Practice: Factoring Trinomials

1. Complete the table for $x^2 + 18x - 40$.

	x	_____
x	x^2	☐ $1x$ ☐ $-1x$ ☐ $8x$ ☐ $-8x$ ☐ $2x$ ☐ $-2x$ ☐ $10x$ ☐ $-10x$ ☐ $4x$ ☐ $-4x$ ☐ $20x$ ☐ $-20x$ ☐ $5x$ ☐ $-5x$ ☐ $40x$ ☐ $-40x$
_____	☐ $1x$ ☐ $-1x$ ☐ $8x$ ☐ $-8x$ ☐ $2x$ ☐ $-2x$ ☐ $10x$ ☐ $-10x$ ☐ $4x$ ☐ $-4x$ ☐ $20x$ ☐ $-20x$ ☐ $5x$ ☐ $-5x$ ☐ $40x$ ☐ $-40x$	-40

2. Complete the table for $x^2 + 12x + 32$.

	x	_____
x	x^2	☐ $1x$ ☐ $-1x$ ☐ $8x$ ☐ $-8x$ ☐ $2x$ ☐ $-2x$ ☐ $16x$ ☐ $-16x$ ☐ $4x$ ☐ $-4x$ ☐ $32x$ ☐ $-32x$
_____	☐ $1x$ ☐ $-1x$ ☐ $8x$ ☐ $-8x$ ☐ $2x$ ☐ $-2x$ ☐ $16x$ ☐ $-16x$ ☐ $4x$ ☐ $-4x$ ☐ $32x$ ☐ $-32x$	32



7.2.4 Your Turn!

1. Complete the table for $x^2 - 11x + 18$.

	x	_____
x	x^2	☐ $1x$ ☐ $-1x$ ☐ $6x$ ☐ $-6x$ ☐ $2x$ ☐ $-2x$ ☐ $9x$ ☐ $-9x$ ☐ $3x$ ☐ $-3x$ ☐ $18x$ ☐ $-18x$
_____	☐ $1x$ ☐ $-1x$ ☐ $6x$ ☐ $-6x$ ☐ $2x$ ☐ $-2x$ ☐ $9x$ ☐ $-9x$ ☐ $3x$ ☐ $-3x$ ☐ $18x$ ☐ $-18x$	18



7.2.5 Wrap-Up: Reflection

1. How “in the zone” do you feel right now about factoring trinomials?



Unit 7, Lesson 3: Factoring a Trinomial with a Leading Coefficient of 1



Benchmark

MA.912.AR.1.7 Rewrite a polynomial expression as a product of polynomials over the real number system.



Learning Target

- ☐ I can factor a trinomial with a leading coefficient of 1.



7.3.1 Warm-Up: Bell Work

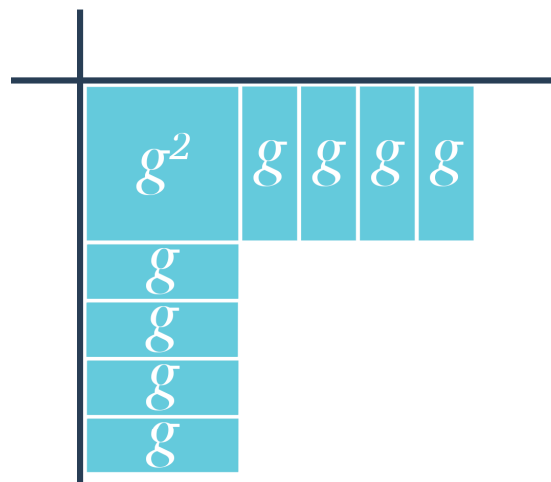
1. Complete the table for $x^2 + 3x - 18$.

	x	_____
x	x^2	☐ $1x$ ☐ $-1x$ ☐ $6x$ ☐ $-6x$ ☐ $2x$ ☐ $-2x$ ☐ $9x$ ☐ $-9x$ ☐ $3x$ ☐ $-3x$ ☐ $18x$ ☐ $-18x$
_____	☐ $1x$ ☐ $-1x$ ☐ $6x$ ☐ $-6x$ ☐ $2x$ ☐ $-2x$ ☐ $9x$ ☐ $-9x$ ☐ $3x$ ☐ $-3x$ ☐ $18x$ ☐ $-18x$	-18



7.3.2 Collaboration: Different Methods of Factoring

1. Garcelle is trying to factor the trinomial $g^2 + 8g + 12$ using algebra tiles.



Discuss with your partner the mistake that Garcelle made. Write a note to Garcelle to explain her mistake and how she can fix it.

Means

I

Start

To

Acquire

Knowledge

Experience

Skills



2. Complete the table for $g^2 + 8g + 12$.

	g	_____
g	g^2	☐ $1g$ ☐ $-1g$ ☐ $4g$ ☐ $-4g$ ☐ $2g$ ☐ $-2g$ ☐ $6g$ ☐ $-6g$ ☐ $3g$ ☐ $-3g$ ☐ $12g$ ☐ $-12g$
_____	☐ $1g$ ☐ $-1g$ ☐ $4g$ ☐ $-4g$ ☐ $2g$ ☐ $-2g$ ☐ $6g$ ☐ $-6g$ ☐ $3g$ ☐ $-3g$ ☐ $12g$ ☐ $-12g$	12

3. What are the two binomial factors for $g^2 + 8g + 12$?

$$g^2 + 8g + 12 = (\quad) (\quad)$$

Felix showed Garcelle how to use the **area model** to factor $g^2 + 8g + 12$.



The **area model** is a rectangular diagram that utilizes the decomposition of side lengths by place value to multiply numbers using the distributive property.



4. Fill in the missing information for the **area model** for $g^2 + 8g + 12$.

	$1g$	6
$1g$		$6g$
2		

5. Garcelle asked Felix how to determine the monomial that belongs in the orange boxes when creating the area model for $k^2 - 3k - 28$. Felix said that he rewrites $k^2 - 3k - 28$ as $k^2 - 7k + 4k - 28$ because $-7 \cdot 4 = -28$ and $-7 + 4 = -3$. He knows that one of the orange boxes should result in the product $-7k$ and the other should result in the product $4k$.

a. Complete the area model for $k^2 - 3k - 28$.

	_____	_____

- b. What are the two binomial factors for $k^2 - 3k - 28$?

$$k^2 - 3k - 28 = (\quad)(\quad)$$

- c. Lori uses a different method of factoring. Like Felix, she rewrote $k^2 - 3k - 28$ as $k^2 - 7k + 4k - 28$, but instead paired the terms as shown.

$$(k^2 - 7k) + (4k - 28)$$

Work with your partner to finish factoring using Lori's method, known as **factoring by grouping**, using the following steps.

$(k^2 - 7k) + (4k - 28)$	
Factor out the greatest common factor from each pair of terms.	$k(k - 7) + \underline{\quad}(k - 7)$
The binomial, $k - 7$, is common so the sum can be rewritten.	$(k - 7)(k + \underline{\quad})$

There are many different methods for factoring trinomials. Relating factoring to algebra tiles or the area model is useful for understanding why the factors of a trinomial of the form $ax^2 + bx + c$ are two binomials.

Using algebra tiles or the area model is also helpful in understanding the relationship the factors of the c -term have with the b -term.

6. Discuss with your partner the relationship the factors of the c -term have with the b -term based on the expressions explored in the previous problems.



7.3.3 Guided Instruction: Factoring Trinomials where $a = 1$



A **quadratic expression**, or **quadratic equation**, is a polynomial expression or equation that contains a term of degree 2, but no term of higher degree.

Factoring quadratic expressions in the form of trinomials such as $ax^2 + bx + c$, where $a = 1$, results in the factors $(x + g)(x + h)$ where, $g \cdot h = c$ and $g + h = b$. Later in this unit, factoring trinomials where a is a value other than 1 will be explored.

1. Consider the trinomials $y^2 + 14y + 24$, $y^2 - 14y + 24$, $y^2 + 10y - 24$, and $y^2 - 10y - 24$.

- a. Factor each trinomial.

$y^2 + 14y + 24$	$y^2 - 14y + 24$	$y^2 + 10y - 24$	$y^2 - 10y - 24$

- b. Describe any similarities between the four trinomials.

- c. Describe any similarities between the factored forms of the four trinomials.

2. Consider the trinomial $x^2 - 2x - 8$.

- a. Factor $x^2 - 2x - 8$ using the area model.

- b. Factor $x^2 - 2x - 8$ using factoring by grouping.

- c. Discuss with your partner how to check that the two binomials of the factored form are correct.



7.3.4 Your Turn!

1. Factor each quadratic expression using the method of your choice.

a. $d^2 + 11d + 24$

b. $a^2 - 9a + 14$

c. $x^2 + 4x - 96$



7.3.5 Wrap-Up: Cool Down

1. Write a short note to a friend who missed class summarizing today's lesson. Include something you learned and a mistake you learned from.

Unit 7, Lesson 4: Factoring a Trinomial with a Leading Coefficient Greater Than 1



Benchmark

MA.912.AR.1.7 Rewrite a polynomial expression as a product of polynomials over the real number system.



Learning Targets

- ☐ I can use algebra tiles to factor a trinomial with a leading coefficient greater than 1.
- ☐ I can factor a trinomial with a leading coefficient greater than 1.



7.4.1 Warm-Up: Growth Mindset

1. Each statement describes the celebrity indicated.

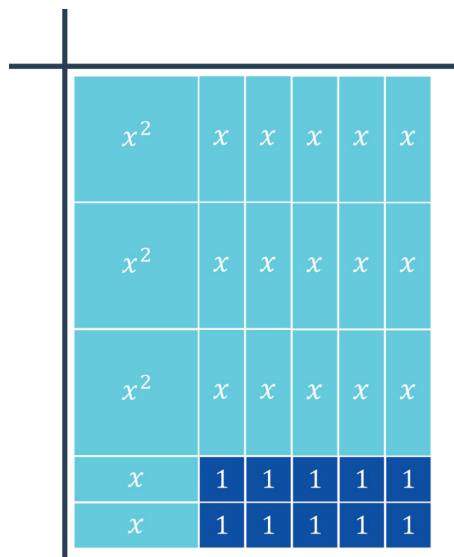
Statement	Celebrity
He was never offered a top Division 1 scholarship. Scouting Reports said his explosiveness and athleticism were below standard.	Steph Curry
His first book was rejected by 27 different publishers. They said his material was too different and would never sell.	Dr. Seuss
He was fired from a newspaper for not being creative enough. They said he lacked imagination and had no good ideas.	Walt Disney

- a. How did a growth mindset help these celebrities?
- b. What can you do to maintain a growth mindset if you get frustrated with today's (or any day's) lesson to help you keep a growth mindset?



7.4.2 Guided Instruction: Factoring Trinomials Where $a \neq 1$

- Use the algebra tiles to determine the two binomials that are factors of $3x^2 + 17x + 10$.



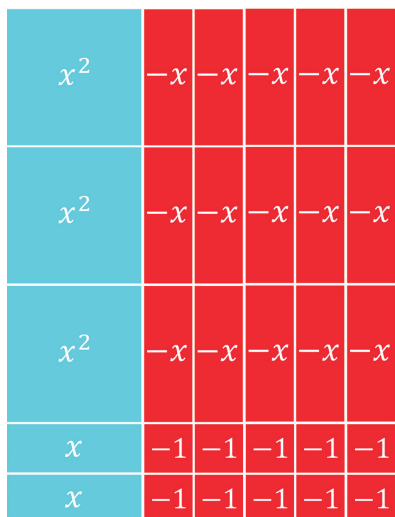
- Discuss with your partner any differences you notice between the binomial factors in question 1 and factoring trinomials where $a = 1$. Summarize your discussion.

- The steps for applying the distributive property to multiply $(3x + 2)(x - 5)$ are shown.

Step	$(3x + 2)(x - 5)$	Result
Distribute $3x$ to x		$3x^2$
Distribute $3x$ to -5		$-15x$
Distribute 2 to x		$2x$
Distribute 2 to -5		-10

- Based on the results in the table, what is the product of $(3x + 2)(x - 5)$?

- b. The algebra tiles diagram models the result of $(3x + 2)(x - 5)$.



Draw a rectangle around $-15x$ in the algebra tiles diagram for $3x^2 - 13x - 10$.

- c. Discuss with your partner how the algebra tiles diagram demonstrates the distribution of $3x$ to -5 . Summarize your discussion.

4. Compare the algebra tiles diagram for $3x^2 + 17x + 10$ to the algebra tiles diagram for $3x^2 - 13x - 10$. Describe the differences.

Factoring trinomials of the form $ax^2 + bx + c$, where $a = 1$, resulted in the factored form $(x + g)(x + h)$, where $g \cdot h = c$ and $g + h = b$.

5. Use the examples in the table to explore factoring trinomials of the form $ax^2 + bx + c$ where $a \neq 1$. Summarize the similarities and differences of the factored forms where $a = 1$ and where $a \neq 1$.

Trinomial	$3x^2 + 17x + 10$	$3x^2 - 13x - 10$
Factored form	$(3x + 2)(x + 5)$	$(3x + 2)(x - 5)$
Similarities		
Differences		

When factoring quadratics of the form $ax^2 + bx + c$, the coefficient b is the sum of factors of the product of a and c .

6. In $3x^2 + 17x + 10$ and $3x^2 - 13x - 10$, the products of a and c are 30 and -30 , respectively.

a. List all the pairs of factors for both 30 and -30 .

Value	Pairs of Factors
30	
-30	

b. Circle the factor pair for 30 whose sum is 17.

c. Circle the factor pair for -30 whose sum is -13 .

d. The factor pairs can be used to factor by grouping. Complete the steps for factoring by grouping for each trinomial expression.

$3x^2 + 17x + 10$	$3x^2 - 13x - 10$
$3x^2 + 2x + 15x + 10$	$3x^2 + 2x - 15x - 10$
$3x(x + \underline{\hspace{1cm}}) + 2(x + \underline{\hspace{1cm}})$	$3x(x - \underline{\hspace{1cm}}) + 2(x - \underline{\hspace{1cm}})$
$(3x + 2)(x + 5)$	$(3x + 2)(x - 5)$



7.4.3 Guided Practice: Factoring Trinomials

Where $a \neq 1$

1. Factor $5x^2 + 6x - 8$ by grouping.

2. Consider the trinomial $3x^2 - 13x - 10$.

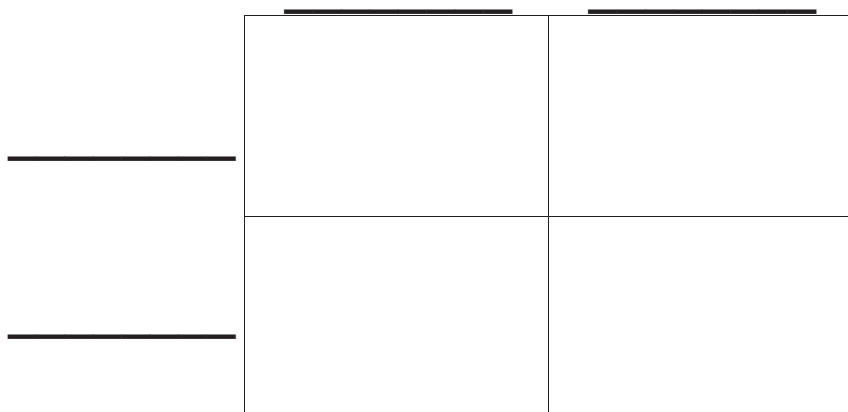
a. Complete the table by choosing the appropriate values.

	$3x$	☐ 1 ☐ 2 ☐ 5 ☐ 10	☐ -1 ☐ -2 ☐ -5 ☐ -10
x	$3x^2$	☐ 1x ☐ 2x ☐ 5x ☐ 10x	☐ -1x ☐ -2x ☐ -5x ☐ -10x
☐ 1 ☐ 2 ☐ 5 ☐ 10	☐ -1 ☐ -2 ☐ -5 ☐ -10	☐ 3x ☐ 6x ☐ 15x ☐ 30x	☐ -3x ☐ -6x ☐ -15x ☐ -30x
-10			

b. Use the values to complete the area model.

	$3x$	
x	$3x^2$	
		-10

3. Factor $5x^2 - 3x - 8$ using the area model.



7.4.4 Try It: Factoring Trinomials Where $a \neq 1$

1. Factor $2z^2 - 21z - 36$.

2. Factor $3b^2 - 23b + 14$.

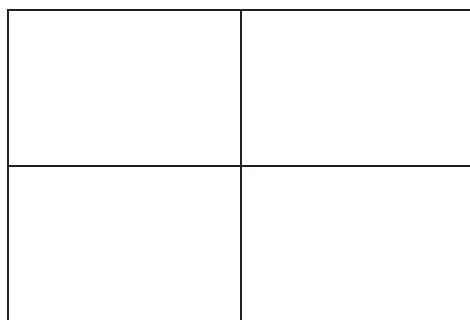


7.4.5 Your Turn!

1. Consider the quadratic expression.

$$5x^2 + 21x + 18$$

- a. Factor using the area model.



- b. Factor by grouping.



7.4.6 Wrap-Up: Reflection

1. What was the **muddiest** (most difficult or confusing) point in today's lesson on factoring trinomials where $a \neq 1$?



Unit 7, Lesson 5: Factoring Trinomials



Benchmark

MA.912.AR.1.7 Rewrite a polynomial expression as a product of polynomials over the real number system.



Learning Target

☐ I can factor trinomials.



7.5.1 Warm-Up: Walking Florida

1. Florida has the longest coastline, 1197 miles, in the contiguous United States. The distance from Key West to the farthest west Florida border on I-10 is 839 miles. The average person walks 3 miles per hour (mph).
 - a. How many hours would it take someone walking 3 mph to walk from Key West to Florida's western border?
 - b. How much longer, in days, would it take someone to walk along Florida's coastline than to walk from Key West to Florida's western border?



7.5.2 Exploration: Relating Terms and Factors

1. The steps for using the distributive property to multiply $(2x + 3)(2x - 2)$ are shown.

Step	$(2x + 3)(2x - 2)$	Result
Distribute $2x$ to $2x$		$4x^2$
Distribute $2x$ to -2		$-4x$
Distribute 3 to $2x$		$6x$
Distribute 3 to -2		-6

- a. Based on the results in the table, what quadratic expression is the product of $(2x + 3)(2x - 2)$?

- b. Tyra and Min-Joon were discussing how to find the x -term of the quadratic expression for $(2x + 3)(2x - 2)$.

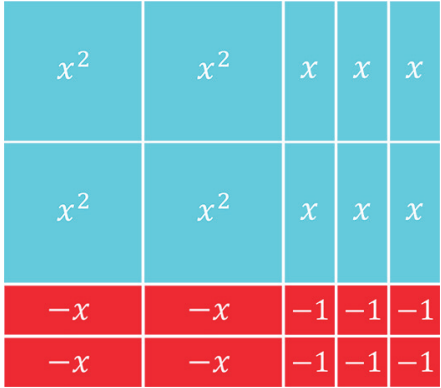
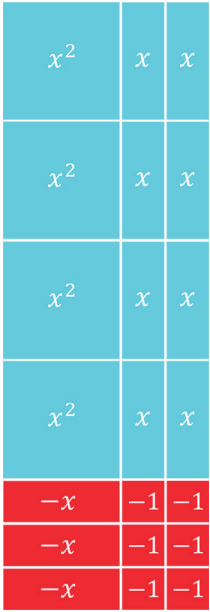
Tyra	Min-Joon
The x -term of $(2x + 3)(2x - 2)$ can be quickly found by adding 3 and -2 .	Looking at the table, the x -term of the quadratic expression is $2x$. I think you could just use what repeats.

Discuss with your partner both students' strategies.

- c. Both students have made a mistake. Describe to Tyra and Min-Joon how to find the x -term of the quadratic expression for $(2x + 3)(2x - 2)$.
- d. Discuss with your partner whether or not the x -term of the quadratic expression $(4x - 3)(x - 2)$ is different from the x -term of the quadratic expression $(2x - 3)(2x - 2)$. Summarize your discussion.

2. Consider the trinomials $4x^2 + 2x - 6$ and $4x^2 + 5x - 6$.

a. Use the algebra tiles to factor each trinomial.

Trinomial	$4x^2 + 2x - 6$	$4x^2 + 5x - 6$
Algebra Tiles		
Factored Form		

b. The trinomials have the same leading term, $4x^2$. List the two ways $4x^2$ is factored.

c. List the factor pairs of -24 .

d. Which pair of factors of -24 add to 2?

e. Discuss with your partner how the pair of factors of -24 that add to 2 is related to the factored form of $4x^2 + 2x - 6$. Show the factors along the algebra tiles.

x^2	x^2	x	x	x
x^2	x^2	x	x	x
$-x$	$-x$	-1	-1	-1
$-x$	$-x$	-1	-1	-1



7.5.3 Activity: Card Sort

1. Work with your partner to sort the cards. Write the letter of the matching card for each representation.

Quadratic Expression	Algebra Tiles	Area Model	Expanded Form	Factored Form
$6x^2 - x - 2$				
$6x^2 + x - 2$				
$6x^2 - 7x + 2$				
$6x^2 - 11x - 2$				
$6x^2 + 11x - 2$				
$6x^2 - 13x + 2$				

2. Describe one strategy you and your partner used to sort the cards.



7.5.4 Guided Instruction: Factoring Trinomials in the Form $ax^2 + bx + c$ when a has Multiple Factors

Factoring trinomials of the form $ax^2 + bx + c$, where the value of a has multiple factors, requires all of the factors of a to be considered when factoring using algebra tiles or the area model.

1. Complete the table for $10x^2 + x - 3$.

	<u> </u> x	<u> </u>
<u> </u> x	$10x^2$	☐ $1x$ ☐ $-1x$ ☐ $6x$ ☐ $-6x$ ☐ $2x$ ☐ $-2x$ ☐ $10x$ ☐ $-10x$ ☐ $3x$ ☐ $-3x$ ☐ $15x$ ☐ $-15x$ ☐ $5x$ ☐ $-5x$ ☐ $30x$ ☐ $-30x$
<u> </u>	☐ $1x$ ☐ $-1x$ ☐ $6x$ ☐ $-6x$ ☐ $2x$ ☐ $-2x$ ☐ $10x$ ☐ $-10x$ ☐ $3x$ ☐ $-3x$ ☐ $15x$ ☐ $-15x$ ☐ $5x$ ☐ $-5x$ ☐ $30x$ ☐ $-30x$	-3

2. Which expanded form of $10x^2 + x - 3$ could be used to factor the expression by grouping?
 - ☐ $10x^2 + 5x - 6x - 3$
 - ☐ $10x^2 - 5x + 6x - 3$
 - ☐ $10x^2 - 10x + 3x - 3$
 - ☐ $10x^2 + 10x - 3x - 3$
3. Use the expanded form to factor $10x^2 + x - 3$.



7.5.5 Wrap-Up: Reflection

1. Complete the table.

The terms and information used in the exploration and activity that I knew were ...	Something I wondered during the exploration or activity was ...	Something I learned during the exploration or activity was ...

Unit 7, Lesson 6: Factoring by Grouping



Benchmark

MA.912.AR.1.7 Rewrite a polynomial expression as a product of polynomials over the real number system.



Learning Target

☐ I can factor a polynomial by grouping.



7.6.1 Warm-Up: Notice and Wonder

1. Coral Castle in Miami, Florida was created by Edward Leedskalnin. He carved over 1100 tons of coral rock using an unknown process, often carving at night.



- a. What do you notice about the sculptures?
- b. What do you wonder about the process Ed used?



7.6.2 Guided Instruction: Factoring by Grouping

When factoring $ax^2 + bx + c$, the coefficient b is the sum of factors of the product of a and c .

1. In both $8x^2 - 19x - 15$ and $8x^2 - 2x - 15$, the product of a and c is -120 .
 - a. List all the pairs of factors for -120 .
 - b. Circle the factor pair for -120 that adds to the sum of -19 .
 - c. Draw a rectangle around the factor pair for -120 that adds to the sum of -2 .

2. The factor pairs can be used to factor by grouping. Complete the steps for factoring by grouping for each trinomial.

$8x^2 - 19x - 15$	$8x^2 - 2x - 15$
$8x^2 - 24x + 5x - 15$	$8x^2 - 12x + 10x - 15$
$8x(x - \underline{\hspace{1cm}}) + 5(x - \underline{\hspace{1cm}})$	$4x(2x - \underline{\hspace{1cm}}) + 5(2x - \underline{\hspace{1cm}})$

Factoring by grouping is helpful when factoring trinomials of the form $ax^2 + bx + c$, where the value of a has multiple factors.

3. Factor $6x^2 - 19x + 15$ by grouping.

Factoring by grouping can also be helpful when factoring a polynomial with four terms.

4. Factor $3y^4 - 6y^3 + 9y^2 - 18y$ by grouping.

5. The polynomial $3y^4 + 9y^2 - 6y^3 - 18y$ is a rearranged form of $3y^4 - 6y^3 + 9y^2 - 18y$.

Factor $3y^4 + 9y^2 - 6y^3 - 18y$ by grouping.

6. What do you notice about the factored forms of $3y^4 - 6y^3 + 9y^2 - 18y$ and $3y^4 + 9y^2 - 6y^3 - 18y$?



7.6.3 Guided Practice: Factoring by Grouping

1. Factor $6x^2 + x - 35$ by grouping.

2. Factor $-15g + 35gh - 7h + 3$ by grouping.



7.6.4 Collaboration: Factoring by Grouping

Work with your partner to complete the following.

1. Factor $2x^2 + 3x + 4x + 12$ by grouping.

2. What happens when you try to group the expression and find a common factor?

3. If you rearrange the middle two terms, what happens when you try to factor by grouping again?

4. Complete the statement.

When factoring an expression by grouping, group terms

that have coefficients that

- ☐ are exactly the same.
- ☐ have common factors.
- ☐ are in numerical order.

5. Work with your partner to create a strategy for the best way to group an expression.



7.6.5 Your Turn!

1. Factor $12x^2 - 16x - 3$ by grouping.



7.6.6 Wrap-Up: Ticket Out the Door

1. Factor each expression.

a. $h^2 - 12h + 27$

b. $36a^4b^7 + 42a^6b^3$

c. $12x^2 - 8x - 15$

d. $5x^2 + 37x + 14$

Unit 7, Lesson 7: Factoring Perfect Square Trinomials



Benchmark

MA.912.AR.1.7 Rewrite a polynomial expression as a product of polynomials over the real number system.



Learning Target

☐ I can factor perfect square trinomials.



7.7.1 Warm-Up: Critical Thinking

1. Fill in the blanks with an integer so that the quadratic expression is factorable. Then, complete the factored form.

Quadratic Expression	Factored Form
$x^2 + \underline{\hspace{1cm}}x + 36$	$(x + \underline{\hspace{1cm}})(x + \underline{\hspace{1cm}})$
$x^2 - \underline{\hspace{1cm}}x - 36$	$(x - \underline{\hspace{1cm}})(x + \underline{\hspace{1cm}})$
$x^2 + \underline{\hspace{1cm}}x - 36$	$(x - \underline{\hspace{1cm}})(x + \underline{\hspace{1cm}})$



7.7.2 Collaboration: Perfect Square Trinomials

1. The quadratic expressions represented by algebra tiles are perfect square trinomials.

A	B

- a. Discuss with your partner why these quadratic expressions are called perfect square trinomials.

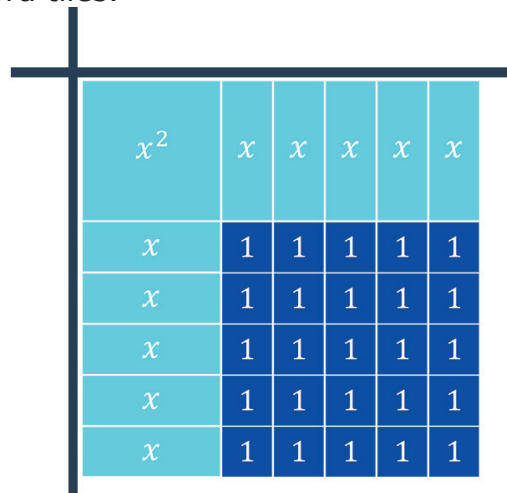
- b. Write the trinomials represented by the algebra tiles.

Trinomial A	Trinomial B

- c. Use the algebra tile representations to write the factored form of each trinomial.

Factored Form of Trinomial A	Factored Form of Trinomial B

2. The quadratic expression $x^2 + 10x + 25$ is represented by the algebra tiles.



- a. Pranjal used the algebra tiles to write the factored form of $x^2 + 10x + 25$ as $(x + 5)^2$.

Check Pranjal's work by completing the steps.

$$(x + 5)^2 = (x + 5)(x + 5)$$

$$(x + 5)(x + 5) = \underline{\hspace{2cm}}$$

- b. Discuss with your partner how writing the factored form of the quadratic expression $x^2 + 10x + 25$ as $(x + 5)^2$ relates to the expression being called a perfect square trinomial. Summarize your discussion.

3. Work with your partner to determine which quadratic expressions are perfect square trinomials.

Expression	Algebra Tiles
$4x^2 - 9$ Is it a perfect square trinomial? ☐ Yes ☐ No	
$4x^2 + 12x + 9$ Is it a perfect square trinomial? ☐ Yes ☐ No	

Expression	Algebra Tiles
$4x^2 - 12x + 9$ Is it a perfect square trinomial? ☐ Yes ☐ No	
$9x^2 + 6x + 1$ Is it a perfect square trinomial? ☐ Yes ☐ No	
$9x^2 - 1$ Is it a perfect square trinomial? ☐ Yes ☐ No	

4. Describe some key characteristics of perfect square trinomials by relating both the representation of a quadratic expression using algebra tiles and the trinomial.



7.7.3 Guided Instruction: Factoring Perfect Square Trinomials

A **perfect square trinomial** is a trinomial of the form $a^2 + 2ab + b^2$ or $a^2 - 2ab + b^2$.

1. The quadratic expression $4x^2 + 12x + 9$ is a perfect square trinomial. Rewrite the trinomial in the form $a^2 + 2ab + b^2$.

Perfect square trinomials of the form $a^2 + 2ab + b^2$ can be factored as $(a + b)^2$.

2. Factor $4x^2 + 12x + 9$.

3. The quadratic expression $9x^2 - 12x + 4$ is a perfect square trinomial. Rewrite the trinomial in the form $a^2 - 2ab + b^2$.

Perfect square trinomials of the form $a^2 - 2ab + b^2$ can be factored as $(a - b)^2$.

4. Factor $9x^2 - 12x + 4$.





7.7.4 Guided Practice: Factoring Perfect Square Trinomials

- Place a check mark in the appropriate columns of the table to indicate whether or not each quadratic expression is a perfect square trinomial. If it is a perfect square, write its factored form as $(a + b)^2$ or $(a - b)^2$.

Quadratic Expression	First Term is a Perfect Square	Third Term is a Perfect Square	Second Term is $2ab$ or $-2ab$	Factored Form
$2g^2 + 6g + 9$				
$16y^2 - 56y + 49$				
$b^2 - 26b + 25$				
$4w^2 - 16w + 8$				
$9x^2 - 4$				
$q^2 - 2qz + z^2$				



7.7.5 Your Turn!

- Factor $16h^2 + 88h + 121$.

- Factor $k^2 - 10k + 25$.



7.7.6 Wrap-Up: Reflection

1. Choose the emoji that reflects your understanding of factoring perfect square trinomials.



0 1 2 3 4 5 6 7 8 9 10

2. Complete the statement.

I chose this emoji because _____



Unit 7, Lesson 8: Factoring the Difference of Two Squares



Benchmark

MA.912.AR.1.7 Rewrite a polynomial expression as a product of polynomials over the real number system.



Learning Targets

- ☐ I can factor a difference of two squares.
- ☐ I can rewrite a polynomial expression as a product of polynomials by factoring.



7.8.1 Warm-Up: Bell Work

1. Factor $36x^2 - 12x + 1$.

2. Factor $9y^2 + 30y + 25$.



7.8.2 Exploration: What is a Difference of Two Squares?

The table shows two algebra tile models and their corresponding expressions.

$4x^2 - 12x + 9$					$4x^2 - 9$				
x^2	x^2	$-x$	$-x$	$-x$	x^2	x^2	x	x	x
x^2	x^2	$-x$	$-x$	$-x$	x^2	x^2	x	x	x
$-x$	$-x$	1	1	1	$-x$	$-x$	-1	-1	-1
$-x$	$-x$	1	1	1	$-x$	$-x$	-1	-1	-1
$-x$	$-x$	1	1	1	$-x$	$-x$	-1	-1	-1

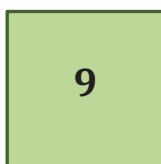
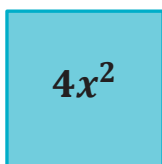
The quadratic expression $4x^2 - 12x + 9$ is a perfect square trinomial.

- Discuss with your partner why the quadratic expression $4x^2 - 9$ is not a perfect square trinomial, even though the algebra tile model is a square. Summarize your discussion.

2. Use the algebra tile models to factor each quadratic expression.

$4x^2 - 12x + 9$	$4x^2 - 9$

3. What do you notice about the two factored forms?
4. The quadratic expression $4x^2 - 9$ is a difference of two squares. Discuss with your partner why $4x^2 - 9$ is called a "difference."
5. The two squares shown represent each term of the binomial $4x^2 - 9$.



What is the side length of each square?

$4x^2$	9
Side Length = _____	Side Length = _____

The squares are a geometric representation of each term. The quadratic expression $4x^2 - 9$ can be rewritten with each term as a monomial squared, as shown.

$$(2x)^2 - (3)^2$$

6. Discuss with your partner how the factored form of $4x^2 - 9$ reflects the expression $(2x)^2 - (3)^2$. Summarize your discussion.



The **difference of two squares** is a binomial of the form $(a)^2 - (b)^2$, where a and b can represent a monomial expression or any value except 0. The factored form is $(a + b)(a - b)$.

7. Use the table to determine if each expression is a difference of two squares.

Quadratic Expression	The Expression is a Binomial that Can Be Written as a Difference	First Term is a Perfect Square	Second Term is a Perfect Square
$-100g^4 + 1$	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
$16y^2 - 100y$	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
$-b^2 + 25$	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
$25w^2 - 40w + 16$	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
$9x^2 - 4y^6$	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
$q^2 - 2qz + z^2$	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
$49g^2 - 81h^2$	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No

- Complete the table.
- Circle the expressions in the table that are considered a difference of two squares, based on your check marks.



7.8.3 Guided Practice: Factoring a Difference of Two Squares

- Consider the expression $49g^2 - 81h^2$.

- Rewrite $49g^2 - 81h^2$ as a difference of two squares.

$$(\quad)^2 - (\quad)^2$$

- Factor $49g^2 - 81h^2$.

- Consider the expression $9x^2 - 4y^6$.

- Rewrite $9x^2 - 4y^6$ as a difference of two squares.

$$(\quad)^2 - (\quad)^2$$

- Factor $9x^2 - 4y^6$.

3. Both $-121y^2 + 81$ and $-1 + 16x^2$ are differences of two squares. Discuss with your partner how to rewrite them as a difference of two squares. Then, work together to factor each.

$-121y^2 + 81$	$-1 + 16x^2$



7.8.4 Try It: Factoring Polynomials

This is a mixed practice, which includes all of the different types of polynomials factored in this unit. If necessary, review previous lessons for examples of how to factor each type.

1. Factor each expression.

a. $49t^2 - 42t + 9$

b. $1 - 25k^2$

c. $36x^2y^7 - 48x^4y^4 + 12x^2y^4$

d. $12k^2 + 3k - 15$

e. $d^2 + d - 42$



7.8.5 Your Turn!

1. Factor each expression.

a. $25m^2 + 20m + 4$

b. $2g^2 + 3g - 20$

c. $y^2 - 5y - 84$

d. $49z^4 - 16w^4$



7.8.6 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each concept.



I need help.



I got it.



I could help someone.

Factoring out a common monomial factor			
Factoring a trinomial where $a = 1$			
Factoring a trinomial where $a > 1$			
Factoring a perfect square trinomial			
Factoring a difference of two squares			